```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

df=pd.read_csv('/content/archive (7).zip')
```

New Section

```python
df.head(), df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1338 non-null   int64  
 1   sex       1338 non-null   object 
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64  
 4   smoker    1338 non-null   object 
 5   region    1338 non-null   object 
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
(   age     sex     bmi  children smoker     region      charges
 0   19  female  27.900         0    yes  southwest  16884.92400
 1   18    male  33.770         1     no  southeast   1725.55230
 2   28    male  33.000         3     no  southeast   4449.46200
 3   33    male  22.705         0     no  northwest  21984.47061
 4   32    male  28.880         0     no  northwest   3866.85520,
 None)
```

```python
total_missing= df.isnull().sum().sum()

missing_per_column = df.isnull().sum()
summary = df.describe(include='all')


total_missing, missing_per_column, summary
```

```
(np.int64(0),
 age         0
 sex         0
 bmi         0
 children    0
 smoker      0
 region      0
```

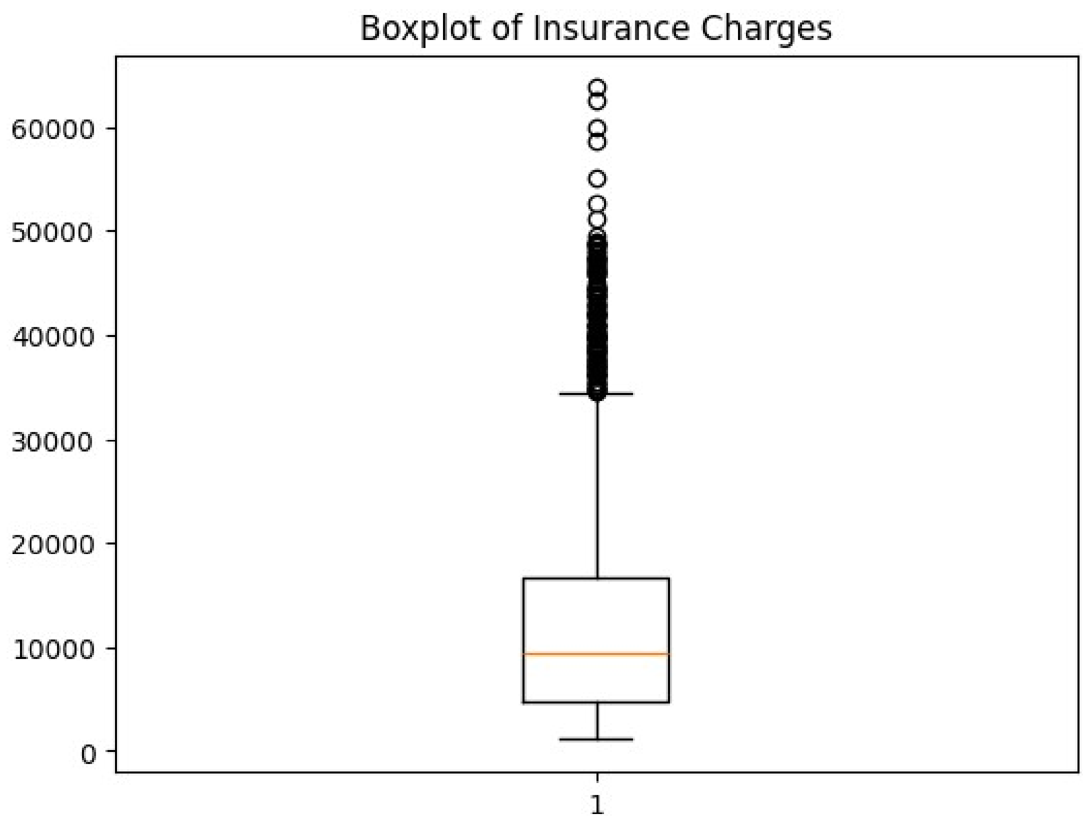
```
 expenses    0
 dtype: int64,
                    age   sex          bmi     children smoker      region
\
 count   1338.000000  1338  1338.000000  1338.000000   1338        1338

 unique          NaN     2          NaN          NaN      2           4

 top             NaN  male          NaN          NaN     no   southeast

 freq            NaN   676          NaN          NaN   1064         364

 mean      39.207025   NaN    30.665471     1.094918    NaN         NaN

 std       14.049960   NaN     6.098382     1.205493    NaN         NaN

 min       18.000000   NaN    16.000000     0.000000    NaN         NaN

 25%       27.000000   NaN    26.300000     0.000000    NaN         NaN

 50%       39.000000   NaN    30.400000     1.000000    NaN         NaN

 75%       51.000000   NaN    34.700000     2.000000    NaN         NaN

 max       64.000000   NaN    53.100000     5.000000    NaN         NaN


             expenses
 count    1338.000000
 unique           NaN
 top              NaN
 freq             NaN
 mean    13270.422414
 std     12110.011240
 min      1121.870000
 25%      4740.287500
 50%      9382.030000
 75%     16639.915000
 max     63770.430000  )
```

```
df_mean = df.copy()
df_mean['bmi'] = df_mean['bmi'].fillna(df_mean['bmi'].mean())

df_median = df.copy()
df_median[' region'] = df_median[' region'].fillna(df_median['
region'].median())
```

```
----------------------------------------------------------------------
-----
KeyError                                  Traceback (most recent call
last)
```

```
/usr/local/lib/python3.12/dist-packages/pandas/core/indexes/base.py in
get_loc(self, key)
   3804         try:
-> 3805             return self._engine.get_loc(casted_key)
   3806         except KeyError as err:

index.pyx in pandas._libs.index.IndexEngine.get_loc()

index.pyx in pandas._libs.index.IndexEngine.get_loc()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

pandas/_libs/hashtable_class_helper.pxi in
pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: ' region'

The above exception was the direct cause of the following exception:

KeyError                                  Traceback (most recent call
last)
/tmp/ipython-input-33161877.py in <cell line: 0>()
      1 df_median = df.copy()
----> 2 df_median[' region'] = df_median[' region'].fillna(df_median['
region'].median())

/usr/local/lib/python3.12/dist-packages/pandas/core/frame.py in
__getitem__(self, key)
   4100             if self.columns.nlevels > 1:
   4101                 return self._getitem_multilevel(key)
-> 4102             indexer = self.columns.get_loc(key)
   4103             if is_integer(indexer):
   4104                 indexer = [indexer]

/usr/local/lib/python3.12/dist-packages/pandas/core/indexes/base.py in
get_loc(self, key)
   3810             ):
   3811                 raise InvalidIndexError(key)
-> 3812             raise KeyError(key) from err
   3813         except TypeError:
   3814             # If we have a listlike key, _check_indexing_error
will raise

KeyError: ' region'

df_mode = df.copy()
df_mode['sex'] = df_mode['sex'].fillna(df_mode['sex'].mode()[0])

df_ffill = df.fillna(method='ffill')
df_bfill = df.fillna(method='bfill')
```

```
/tmp/ipython-input-2683954772.py:1: FutureWarning: DataFrame.fillna
with 'method' is deprecated and will raise in a future version. Use
obj.ffill() or obj.bfill() instead.
  df_ffill = df.fillna(method='ffill')
/tmp/ipython-input-2683954772.py:2: FutureWarning: DataFrame.fillna
with 'method' is deprecated and will raise in a future version. Use
obj.ffill() or obj.bfill() instead.
  df_bfill = df.fillna(method='bfill')


numeric_cols = ['age', 'bmi', 'children', 'charges']

plt.figure()
plt.boxplot(df['charges'])
plt.title("Boxplot of Insurance Charges")
plt.show()
```




```
Q1 = df[numeric_cols].quantile(0.25)
Q3 = df[numeric_cols].quantile(0.75)
IQR = Q3 - Q1
outliers_iqr = ((df[numeric_cols] < (Q1 - 1.5 * IQR)) |
(df[numeric_cols] > (Q3 + 1.5 * IQR))).sum()
```

```python
z_scores = np.abs(stats.zscore(df[numeric_cols]))
outliers_z = (z_scores > 3).sum(axis=0)

lower = df[numeric_cols].quantile(0.01)
upper = df[numeric_cols].quantile(0.99)
outliers_percentile = ((df[numeric_cols] < lower) | (df[numeric_cols]
> upper)).sum()

outliers_iqr, outliers_z, outliers_percentile
```

```
(age           0
 bmi           9
 children      0
 charges     139
 dtype: int64,
 array([ 0,  4, 18,  7]),
 age          0
 bmi         28
 children     0
 charges     28
 dtype: int64)
```